

Lab 2 Draft - Team Sapphire
Zachary Boudreaux
CS 411W
Professor T. Kennedy
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Table of Contents

1 Introduction 3

1.1 Purpose 3

1.2 Scope 3

1.3 Definitions, Acronyms, and Abbreviations..... 4

1.4 References..... 4

1.5 Overview 5

2 Overall Description..... 6

2.1 Product Perspective 6

2.2 Product Functions 6

2.3 User Characteristics..... 7

2.4 Constraints 7

2.5 Assumptions and Dependencies..... 7

1 Introduction

This section provides an introduction of the Software Requirements Specification (SRS) document. It defines the purpose and scope of the Sapphire Sounds project, and identifies the software we are delivering. It includes the following subsections: Purpose, Scope, Definitions, Acronyms and Abbreviations, References and Overview.

1.1 Purpose

The purpose of this Software Requirements Specification is to define the functional and non-functional requirements of the Sapphire Sound system. This document serves as a formal reference for developers, and evaluators, ensuring a shared understanding of what the product is expected to accomplish. It provides a clear description of the system's intended functions, user characteristics, and operating environment to guide design, implementation and testing efforts.

1.2 Scope

Sapphire Sound is a web-based noise monitoring solution designed to provide objective, privacy-preserving data on sound disturbances in shared living environments. The system bridges the gap between subjective noise complaints and verifiable evidence by combining real-time noise monitoring with timestamped event reporting. By doing so, Sapphire Sound enhances tenant accountability, assists property managers in fair dispute resolution, and fosters more peaceful community living. Its benefits include reduced tenant conflicts, improved management transparency, and greater trust between residents and property owners.

1.3 Definitions, Acronyms, and Abbreviations

- **dB (Decibel):** A unit to measure sound intensity
- **Noise-detection Sensor:** A physical device that monitors sound levels without recording audio content
- **Noise Event:** An occurrence when decibel thresholds are reached for a specific duration, triggering a reportable incident
- **Noise Report:** A structured record documenting the time and duration of a noise event
- **Tenant:** A resident or occupant of a shared or multi-unit housing space using the system to monitor and manage noise activity

1.4 References

Team Sapphire. (2025, September 23). *Lab 1 – Sapphire Sound*.

<https://zboudreaux99.github.io/Sapphire-Sound-Monitoring-Website/labs/>

American Public Health Association. (2021, October 25). *Noise as a public health hazard*.

<https://www.apha.org/policy-and-advocacy/public-health-policy-briefs/policy-database/2022/01/07/noise-as-a-public-health-hazard>

Hahad, O., Kuntic, M., Al-Kindi, S. *et al.* Noise and mental health: evidence, mechanisms, and consequences. *J Expo Sci Environ Epidemiol* **35**, 16–23 (2025).

<https://www.nature.com/articles/s41370-024-00642-5>

Munzel, T., Molitor, M., Kuntic, M., et al. (2024, April 25). *Transportation Noise Pollution and Cardiovascular Health*.

<https://www.ahajournals.org/doi/10.1161/CIRCRESAHA.123.323584>

1.5 Overview

The remainder of this document provides a detailed description of the Sapphire Sound system.

Section 2 gives an overview of the product's architecture, key features, and intended users.

Later sections will describe specific functional requirements, performance criteria, and design constraints in accordance with IEEE 830-1998 standards.

2 Overall Description

This portion of the Software Requirements Specification has detailed descriptions of the product and its requirements. It is intended to provide context and clarity for developers and evaluators on the product's expected role.

2.1 Product Perspective

Sapphire Sound is a standalone web application integrated with a compact hardware sensor. The system collects sound intensity data from a decibel-measurement device connected to a Raspberry Pi Zero 2 W controller. The data is transmitted securely to a backend service for analysis and logging. Users access the application through an intuitive React-based frontend, where tenants can monitor their own noise activity and property managers can review timestamped noise reports. This architecture provides both real-time feedback and long-term documentation, making Sapphire Sound a privacy-centric alternative to traditional compliance systems that are invasive and feel like surveillance.

2.2 Product Functions

Sapphire Sound provides many functions, the primary ones being real-time noise monitoring, noise event reporting, tenant and manager dashboards, privacy-first data collection, reward system. The real-time noise monitoring feature persistently measures ambient sound intensity and detects when it exceeds configured thresholds. The noise event reporting feature automatically generates timestamped reports when excessive noise persists beyond the set duration. The tenant and manager dashboards provide a personal view at the details you should see dependent on your role; tenants will see personal data while managers will see data for their tenants. Privacy-first data collection ensures no audio content is recorded or analyzed, only

decibels levels are collected. Lastly, the reward system feature will encourage tenants to participate and remain compliant through incentives for sustained acceptable noise levels.

2.3 User Characteristics

Sapphire Sound users will fall into two buckets: tenant and property managers. Tenants are simply the individuals living in shared housing who utilize the system to monitor their personal noise levels, receive alerts, and view historical data. The technical skill required of tenants is minimal. The property managers are administrative users responsible for reviewing noise event reports, resolving disputes and maintaining community standards. The technical skill required of property managers is moderate, as they will need basic web management tool experience or familiarity.

2.4 Constraints

The system does run into some constraints, namely data retention and privacy compliance. The data retention becomes a storage issue, dependent on how long you plan to store historical data. Assuming you would take a cloud approach, this would require consistent network connectivity to send data to cloud storage for long term storage. The system must comply with privacy and tenant-protection regulations, ensuring no identifiable audio content is recorded or stored. These are both certainly not blocking constraints, but they do introduce increased complexity.

2.5 Assumptions and Dependencies

The assumptions the Sapphire Sound relies on are tenants and property managers have access to internet-enabled devices such as computers, noise sensors are properly installed and maintained, and users will participate in the system to ensure it succeeds. The dependencies

that Sapphire Sound will require is reliable cloud infrastructure for data storage, and third-party software frameworks such as React.js and Node.js.